

Operations Research I

Review Questions: Simplex & Duality

IE 311

A simplex tableau:

z	x_1	x_2	x_3	x_4	x_5	RHS
1	-3	0	0	$\frac{5}{2}$	0	30
0	1	0	1	0	0	4
0	0	1	0	$\frac{1}{2}$	0	6
0	3	0	0	-1	1	6

An LP given in *unstructured* form – it is not directly in the standard maximization form with an obvious initial basic feasible solution. The next slide states such an LP explicitly.

$$\begin{array}{ll}\min & x_0 = x_1 + x_2 \\ \text{s.t.} & x_1 + x_2 \geq 4, \\ & x_1 + 3x_2 \leq 6, \\ & x_1, x_2 \geq 0.\end{array}$$

$$\begin{array}{ll}\max & x_0 = 6x_1 + 3x_2 + 2x_3 \\ \text{s.t.} & 8x_1 + 6x_2 + x_3 \leq 48, \\ & 4x_1 + 2x_2 + \frac{3}{2}x_3 \leq 20, \\ & 2x_1 + \frac{3}{2}x_2 + \frac{1}{2}x_3 \leq 8, \\ & x_1, x_2, x_3 \geq 0.\end{array}$$

Suppose x_1 , x_4 , and x_5 are in the basis.

$$\begin{array}{ll}\max & 2x_1 + 3x_2 + 6x_3 \\ \text{s.t.} & 2x_1 + 5x_2 + x_3 \leq 14, \\ & x_1 - x_2 + x_3 \geq 8, \\ & x_1, x_3 \geq 0, ; x_2 \leq 0.\end{array}$$

Is $(0, 0, 8)$ optimal?